

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 13

DISTRIBUTION (FACTORS PRICING)

DISTRIBUTION AMONG FACTORS:

National wealth / Income is the result of combined efforts of four factors of production e.g. Land, Labour Capital and Organization. Thus every factor gets its due share out of the National Income produced by it. The share of each factor is regarded as.

Factor:	Land	2- Labour	3. Capital	4, Organization
	↓	↓	↓	↓
Share:	Rent	Wages	Interest	Profit

The question arises that how these rewards are given to the factors. What is the principle or formula under which National Income / Wealth is distributed among these factors. In this respect following two theories are given due importance.

1. MARGINAL PRODUCTIVITY THEORY OF RESOURCE PRICING

Marginal productivity theory was originally introduced by classical Economists, most important being Alfred Marshall and I.B. Clark. It provides basis for the distribution of National Income among factors. The principle applied is that "the price or reward" of any factor of production depends upon its marginal product. And the level of output at which it is determined is where MRP (Marginal Revenue Product) = MFC (Marginal Factor Cost paid to the factors). Simply we can say that each factor gets its share equal to its Marginal product.

While by "Marginal Product is meant addition in the total product due to introduction of an additional or extra unit of any variable factor in the production process."

Normally, Marginal product is the product of the last variable unit applied in the business because its Marginal Product Revenue (MRP) = Marginal Factor Cost (MFC).

It is further explained keeping in view the following facts and figures.

1. Only the units of labour (L) are applied one by one as a variable factor.
2. Wage rate is Rs.500/- per week per labourer.
3. Market price per unit output is Rs.50/-

The mechanism of labour pricing is shown as under with the help of a schedule

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

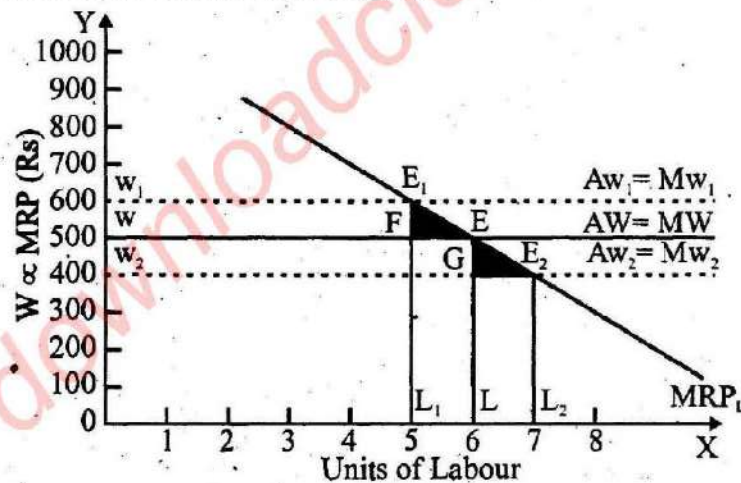
Units of Labour (L)	Total Product (TP)	Marginal Product (MP)	Marginal Product Revenue (MRP) MP (Rs.50/-)	(MFC) = Wage Rate (Rs) per week
1	20	20	20(50) = 1000	500
2	38	18	18(50) = 900	500
3	54	16	16(50) = 800	500
4	68	14	14(50) = 700	500
5	80	12	12(50) = 600	500
6	90	10	10(50) = 500	500
7	98	8	8(50) = 400	500

EXPLANATION:

In the above table $MRP > \text{Wage rate (MFC)}$ upto 5th unit of labour applied in the business. It is profitable for the firm to go on employing more units of labour. But 6th units MRP (500/-) is equal to wage rate (Rs.500/-). Thus 6th will be the last (Marginal) unit in the business. As it's $MRP = \text{Wage rate}$ (also known as marginal factor cost) $MFC = 500/-$ thus we can conclude that every unit of labour is getting its share equal to its marginal product or in money terms = MRP (Rs.500/-)

If the firm employs 7th unit its' $MRP < W$ as (Rs.400 < Rs.500). Thus it will give loss to the firm which is $500 - 400 = 100$ (Rs.). Hence we can say that the firm, to maximize profit, must employ 6 units of labour. If she employs more or less, her profit will decrease which will be unacceptable to her.

This situation is also shown in the diagram.



$AW = MW$ is the Average and Marginal wage rate curve. It remains constant and parallel to x-axis because the firm is functioning under perfect competition. Thus the firm can employ every additional unit at the market curve. It is while MRP is the marginal revenue product curve. It is negatively sloped due to the application of law of diminishing returns. $MW = MRP$ at point E, which is the equilibrium point and shows maximum profit of the firm, when it hires OL (6) unit of labour. If the firm hires less units of labour say OL_1 (5), the profit will be decreased shown in the diagram by FEE_1 (shaded area). On the other side. If firm hires more units of labour as OL_2 (7) again its profit will decrease by EGE_2 (shaded area).

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Note: [MRP, curve is the firms demand curve for Labour].

From the above analysis the final result can be concluded in the form "Marginal Productivity equation "OR" Proportionality Rule."

"When many factors are employed e.g. N.L.K.O; OR (a, b,c,....n) every factor's units are used according to the following equation:-

$$\frac{MPP_a}{P_a} = \frac{MPP_b}{P_b} = \frac{MPP_c}{P_c} = \dots\dots\dots = \frac{MPP_n}{P_n}$$

Here (i) MPP = Marginal Physical Product

(ii) a, b,c,n are the factors of production employed.

(iii) Pa, Pb, Pc,.....Pn is the reward or price paid to each factor.

The above equation shows that every factors $\frac{MPP}{P}$ is equal to the other:

NUMERICAL EXAMPLE:

Note (Here every factors MPP is measured is K.G and price (P) in Rupees)

Land (N)	Labour (L)	Capital (K)	Organization (O)
$\frac{30}{10}$	$\frac{15}{5}$	$\frac{24}{8}$	$\frac{45}{15}$
3	3	3	3

ASSUMPTIONS OF THEORY

Marginal Productivity Theory is based upon following assumptions.

1. Homogeneity in all the units of a factor.
2. Perfect competition in factors as well as product markets.
3. Substitutability in the units of factors of production.
4. Perfect mobility of factors of production from one place to other place or from one profession to the other for higher rewards and other benefits.
5. Elastic supply of factors.
6. Application of the "law of decreasing Returns" on production process in each sector of the economy.

CRITICAL ANALYSIS OF THE THEORY

This classical theory of distribution has been criticized on the following grounds.

1. **ASSUMPTIONS ARE UN-REALISTIC AND INAPPLICABLE IN REAL LIFE.**
 - (a) All the units of a factor are hardly Homogenous. Different pieces of land differ in fertility. All the labourers cannot be equally educated, experienced, efficient and able. Efficiency of capital goods also differs. Entrepreneurs are not equal in ability knowhow, skill, education, training, experience etc.
 - (b) Perfect mobility of factors is not possible e.g. Land cannot be interchanged. Moreover there are some social, cultural, economic and political hurdles in the mobility of labour, Entrepreneur etc.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

- =====
- (c) Factors are not perfect substitutes for each other. Sometime technically it becomes impossible to substitute Labour for capital or land for labour etc.
 - (d) Conditions of perfect competition are hardly fulfilled in the factor or product market.
 - (e) It is not necessary that "Law of decreasing Returns" is always in operation with respect to productive activities. Some producing units may be functioning under law of increasing or constant returns.

2. DIFFICULTY IN FINDING OUT MARGINAL PRODUCT:

According to Prof. Taussig and Devon Port Total Production in the result of joint efforts of all the required factors. Thus it becomes very difficult to determine marginal product of a single factor separately.

3. LOSS OF WITHDRAWAL OF ONE UNIT OF A FACTOR:

Prof. Hobson says, "If a unit of a factor is withdrawn from a production plant or business, the whole production process is disturbed. Under such an up-set situation the marginal product is not reliable or meaningful.

4. FACTORS SUPPLY IS IGNORED:

This theory stresses upon the demand for factors while it ignores the supply conditions of factors. But in reality both sides (demand and supply) should be given equal importance with respect to determine. Marginal product and price determination of a factor.

5. DIFFICULTY IN EVALUATING MARGINAL PRODUCT (MRP)

Some people are simultaneously involved in multi purpose activities e.g. Agriculture, Poultry, fish farming, sheep, goats or dairy farming etc. It becomes very difficult to find out the marginal product from every activity. Moreover, taxes are included in market prices. Sometimes subsidies are given. Thus the real market price upon which MRP depends is very difficult to be identified.

6. INTERDEPENDENCE

IN reality both the marginal product of a factor and it's reward are dependent upon each other. These can't be regarded independent as assumed in the theory. Such as Marginal Product affects wages, while on the other hand wage rate also influences the productivity of the labourers. Hobson, Robinson, Hicks, Samuelson and wick-stead are some big names who criticized this theory while pointing out its weaknesses.

MODERN THEORY OF FACTORS PRICING (DEMAND AND SUPPLY THEORY)

According to the Modern Theory the rewards of factors of production are governed by their demand and supply forces, just as the prices of goods are determined by the equilibrium point of their demand and supply. The only difference is that the demand for goods is direct while demand for factors is "derived" and "indirect."

By the demand of factors we mean the number of units of a factor a firm requires to put into business. While their supply shows the availability of factor units to be employed.

DEMAND FOR FACTORS

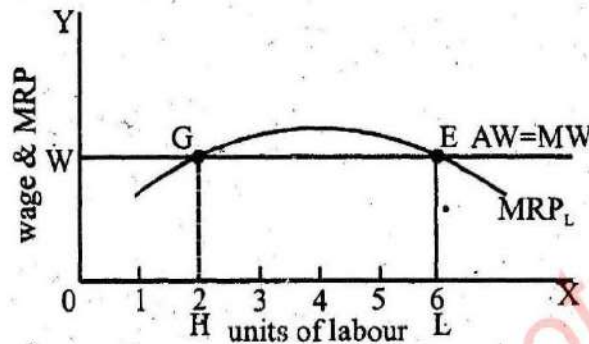
It can be explained as following:

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

(A) UNDER PERFECT COMPETITION:

When all the conditions of perfect competition are fulfilled in the market, No firm can influence the price of a factor by her own individual action.

Thus the units of a factor are employed upto the level where its Marginal Product becomes equal to its wage rate, it is clear from the following diagram.



EXPLANATION:

In the diagram in order to find out the demand for labour, units of labour (L) are shown along x-axis while wages and MRP are measured along y-axis. Due to perfect competition $AW=MW$ (Parallel to x-axis) MRP_L initially rises and then start falling. Both these curves intersect each other at two points (G and E). After point G, MRP_L is still rising which means that additional unit of labour, if employed will increase MRP_L and it will be higher than wage rate resulting a rise in profit. While after point E, MRP_L of additional labour unit is falling and it becomes lower than wage rate. It will cause loss to the firm. Thus "E" is the equilibrium point. Where $MRP_L = MW = AW$. At this point real wage rate is OW and OL(6) units of labour are demanded.

UNDER MONOPOLY OR IMPERFECT COMPETITION

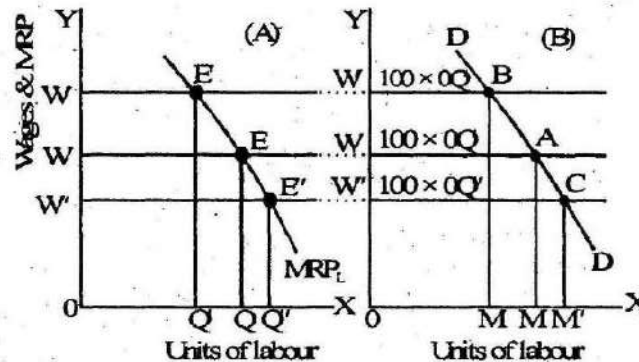
Under this situation wage rate can be influenced by the individual behaviour of the firm. Under Monopoly as only one firm demands the units of factors, thus with the increase in the demand for labour, wage rate may go up, while a decrease in labour demand, wages may also decrease.

As a monopolistic firm represents the whole industry thus, there is only one demand curve for the both.

MARKET DEMAND CURVE

Under perfect competition there is a large number of firms functioning in an industry. Thus in view of the number of firms labour demand curve for an industry may be derived as under:

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)



EXPLANATION:

Diagram (A) shows three levels of wages such as point E, E' and E'' as MRP_L intersects wage curves at these points. Thus the part of MRP_L from E' to E'' is the demand curve of labour for a firm.

In diagram (B) it is assumed that the industry consists upon 100 firms. Thus demand for the industry is 100 times more than the demand for a firm, hence $OM = OQ \times 100$. With the increase in wage rate demand for labour falls and vice versa.

Thus D_L (demand for labour) is a decreasing function of real wage rate $\left(\frac{W}{P}\right) = w$.

Here D_L demand for labour W = nominal wages, P = Price level, w = real wages. Thus DD (D_L) is negatively sloped.

Note: Just like labour demand curve, other factors demand curves may also be drawn.

SUPPLY OF FACTORS

Supply of a factor depends upon many things. For instance, labour supply (S_L) depends upon the number of labour force it's age, sex, literacy rate and population growth rate etc. Except these factors S_L mainly depends upon real wage rate (w). There is an increasing functional relationship between S_L and w . And $S_L = f\left(\frac{W}{P}\right) = w$.

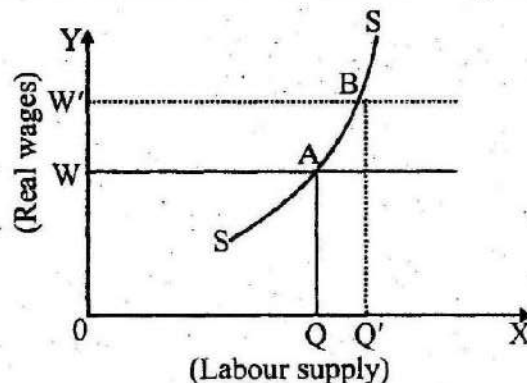
SL = Labour supply,

W = Nominal wages

P = Price level,

w = Real wages

Labour supply curve S_s has been drawn in the following diagram.



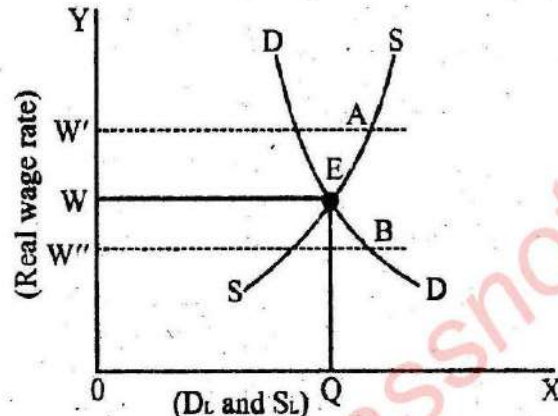
ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

EXPLANATION:

At OW wage level OQ is the labour supply. When real wage rate increases to OW', labour supply rises to OQ'. Thus SS labour supply curve moves upward from left to right showing positive slope.

FACTORS PRICING DETERMINATION

Keeping in view the labour supply and demand forces wage rate is determined at the equilibrium level E as shown in the following diagram.



EXPLANATION:

D_L and S_L have been shown along x-axis, while real wage rate (w) along y-axis. DD is the labour demand curve while Ss is the labour supply curve. Both these curves intersect each other at point E which shows OQ equ-quantity and OW wage rate. If wage rate increases from OW to OW' , labour supply will increase and wage rate will come down to OW on the other hand if, wage rate falls from ow to ow'' , demand for labour will increase and wages will rise upto OW . Thus OW wage rate will be determined and remain stable.

Note: [Other factors reward is also determined by their demand and supply forces].

OBJECTIVE QUESTIONS

Select and tick (✓) correct answer:

- What is the relationship between AW and MW under perfect competition:
 (a) $AW > MW$ (b) $AW < MW$ (c) $AW = MW$ (d) No relationship
- When $\frac{MPP}{P}$ of every factor is equal such combination of factors shows:
 (a) Minimum profit (b) Loss (c) Maximum profit (d) No profit
- Under Marginal productivity theory:
 (a) Units of different factors can be substituted with each other.
 (b) Units of factors are not substitutable.
 (c) No Mobility of factor units
- Who said, "If we withdraw one unit of any factor from working business, whole of the system will collapse."
 (a) Samuelson (b) Wickstead (c) Hobson (d) Robinson

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

5. Which of following has no "derived demand" for the construction of a house:
(a) Bricks (b) Cement (c) Labour (d) Cooking Range
6. Who presented "Classical Marginal Productivity Theory" of factors pricing.
(a) Adam Smith and Ricardo (b) Alfred Marshall and J. B. Clark
(c) Johnson and Johnson (d) Hicks and Hanson
7. Which one of the following is an increasing function of real wages (W).
(a) Demand for labour (D_L) (b) Rate of interest (i)
(c) Supply of labour (S_L) (d) Investment (I)
8. Under perfect competition MRP is sloped:
(a) Positively (b) Negatively (c) Rising but flatter (d) Horizontal
9. How demand for labour (D_L) is influenced by an increase in wage rate:
(a) Rises (b) No change (c) No relationship (d) falls
10. How factors pricing are determined under modern theory.
(a) Like Inferior goods (b) Demand and supply of equilibrium of factors
(c) Determined by the government
(d) According to the demands of trade union.

DESCRIPTIVE PART - I

SHORT NOTES

1. Bring out four assumptions of the marginal Productivity theory?

- (i) Homogeneity in all the units of factor
- (ii) Perfect competition in factor as well as product market.
- (iii) Substitutability and mobility of factors of production.

2. Point out three weaknesses of marginal productivity theory?

- (i) Assumptions are un-realistic
- (ii) Difficulty in finding out marginal product
- (iii) Factors supply is ignored.

3. How factors pricing are determined under modern theory?

According to the modern theory factors pricing are determined through the interaction of forces of demand for and supply of the factors of production. It is the same process as that of the determination of the prices of the goods in a market.

4. What are the determinates of labour supply?

Labour supply depends upon:

- (i) The number of labour force
- (ii) It's age, sex, literacy rate and population growth rate etc
- (iii) But mainly it depends upon real wage rate.

5. What is the "message" given by marginal productivity theory?

According to this theory the price or reward of any factor of production depends upon its marginal product. Production is determined where

$MRP = W$ (Marginal revenue product = wages)

PART - II

1. Explain "Marginal Productivity Theory".
2. Critically examine "classical theory" of factors pricing.
3. Explain "Modern theory" of distribution.